Primary Drivers of Complete Response Letters
Distribution of Deficiency Domains

deficiency_domain				
deficiency_domain	**Frequency**	**Percent**	**Cumulative Frequency**	**Cumulative Percent**
CMC	314	51.48	314	51.48
Clinical	147	24.10	461	75.57
Regulatory	64	10.49	525	86.07
Nonclinical	61	10.00	586	96.07
Statistics	13	2.13	599	98.20
Patent Certifications	11	1.80	610	100.00

Primary Drivers of Complete Response Letters
Deficiency Domains by Application Type (NDA vs BLA)

Frequency
Row Pct

Table of application_type by deficiency_domain

application_type(application_type)	deficiency_domain(deficiency_domain)						
	CMC	Clinical	Regulatory	Nonclinical	Statistics	Patent Certifications	Total
NDA	233 47.26	125 25.35	54 10.95	60 12.17	10 2.03	11 2.23	493
BLA	81 69.23	22 18.80	10 8.55	1 0.85	3 2.56	0 0.00	117
Total	314	147	64	61	13	11	610

Statistics for Table of application_type by deficiency_domain

Statistic	DF	Value	Prob
Chi-Square	5	25.9178	<.0001
Likelihood Ratio Chi-Square	5	34.0066	<.0001
Mantel-Haenszel Chi-Square	1	18.6411	<.0001
Phi Coefficient		0.2061	
Contingency Coefficient		0.2019	
Cramer's V		0.2061	

Sample Size = 610

Primary Drivers of Complete Response Letters
Clinical Deficiency Subtypes

Frequency
Row Pct

Table of deficiency_domain by deficiency_subtype

deficiency_domain(deficiency_domain)	deficiency_subtype(deficiency_subtype)							
	Efficacy	Safety	Human Factors	Pharmacology	Study Design	Risk-Benefit	Data Integrity	Total
Clinical	63 31.03	49 24.14	31 15.27	25 12.32	16 7.88	10 4.93	9 4.43	203
Total	63	49	31	25	16	10	9	203

Frequency
Row Pct

Table of deficiency_domain by deficiency_subtype

deficiency_domain(deficiency_domain)	deficiency_subtype(deficiency_subtype)				
	Facility	Specifications	Manufacturing Controls	Redacted	Stability
CMC	232 42.26	81 14.75	79 14.39	55 10.02	51 9.29
Total	232	81	79	55	51

Table of deficiency_domain by deficiency_subtype

deficiency_domain(deficiency_domain)	deficiency_subtype(deficiency_subtype)		
	Microbiology	Biopharmaceutics	Total
CMC	34 6.19	17 3.10	549
Total	34	17	549

Frequency
Row Pct

Table of deficiency_domain by deficiency_subtype

deficiency_domain(deficiency_domain)	deficiency_subtype(deficiency_subtype)				
	Insufficient Nonclinical Support	Toxicology	Redacted	Impurity Qualification	Total
Nonclinical	20 29.41	20 29.41	16 23.53	12 17.65	68
Total	20	20	16	12	68

Frequency
Row Pct

Table of deficiency_domain by deficiency_subtype

deficiency_domain(deficiency_domain)	deficiency_subtype(deficiency_subtype)					
	Labeling	Administrative/Filing	Redacted	REMs/PMRs	PREA	Total
Regulatory	47 68.12	14 20.29	4 5.80	3 4.35	1 1.45	69
Total	47	14	4	3	1	69

Primary Drivers of Complete Response Letters
Co-occurrence of Deficiency Domains

pattern	COUNT	PERCENT
CMC	172	42.8928
CMC, Clinical	51	12.7182
Clinical	49	12.2195
CMC, Regulatory	27	6.7332
CMC, Nonclinical	21	5.2369
CMC, Clinical, Nonclinical	11	2.7431
Nonclinical	11	2.7431
CMC, Clinical, Regulatory	10	2.4938
Regulatory	8	1.9950
Clinical, Statistical	7	1.7456
CMC, Regulatory, Patent	6	1.4963
Clinical, Nonclinical	6	1.4963

cmc_pattern	COUNT	PERCENT
Facility	135	42.9936
Redacted	24	7.6433
Specifications	19	6.0510
Facility, Redacted	18	5.7325
Facility, Manufacturing Controls	11	3.5032
Manufacturing Controls	10	3.1847
Facility, Specifications, Manufacturing Controls, Stability	9	2.8662
Facility, Specifications	7	2.2293
Facility, Specifications, Manufacturing Controls, Stability, Microbiology	7	2.2293
Facility, Specifications, Stability	7	2.2293
Specifications, Manufacturing Controls, Stability	7	2.2293
Biopharmaceutics	6	1.9108
Facility, Microbiology	5	1.5924

Model Information	
Data Set	SASDATA.DOMAIN_EVENT_MODEL
Response Variable	cmc_f
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	401
Number of Observations Used	401

Logistic Regression: Predicting CMC Deficiencies

Response Profile		
Ordered Value	cmc_f	Total Frequency
1	0	87
2	1	314

Probability modeled is cmc_f=1.

Class Level Information		
Class	Value	Design Variables
application_type	BLA	1
	NDA	-1

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	421.470	419.685
SC	425.464	427.673
-2 Log L	419.470	415.685

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	3.7849	1	0.0517
Score	3.5487	1	0.0596
Wald	3.4767	1	0.0622

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
application_type	1	3.4767	0.0622

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	1.4580	0.1595	83.5543	<.0001
application_type	BLA	1	0.2974	0.1595	3.4767	0.0622

Logistic Regression: Predicting CMC Deficiencies

Odds Ratio Estimates			
Effect	**Point Estimate**	**95% Wald Confidence Limits**	
application_type BLA vs NDA	1.813	0.970	3.387

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	21.6	**Somers' D**	0.097
Percent Discordant	11.9	**Gamma**	0.289
Percent Tied	66.4	**Tau-a**	0.033
Pairs	27318	**c**	0.549

Model Information	
Data Set	SASDATA.DOMAIN_EVENT_MODEL
Response Variable	cmc_f
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	401
Number of Observations Used	401

Response Profile		
Ordered Value	**cmc_f**	**Total Frequency**
1	1	314
2	0	87

Probability modeled is cmc_f=1.

Class Level Information					
Class	**Value**	**Design Variables**			
application_type	**BLA**	1			
	NDA	-1			
regulatory_pathway	**351(a)**	1	0	0	0
	351(k)	0	1	0	0
	505(b)	0	0	1	0
	505(b)(2)	-1	-1	-1	-1
	Not Available	0	0	0	1

Logistic Regression: Predicting CMC Deficiencies
Predict CMC with Multiple Predictors

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	421.470	414.294
SC	425.464	442.251
-2 Log L	419.470	400.294

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	19.1765	6	0.0039
Score	18.9188	6	0.0043
Wald	16.5099	6	0.0113

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
application_type	1	0.4327	0.5106
year	1	1.3318	0.2485
regulatory_pathway	4	14.1367	0.0069

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-95.6279	84.2441	1.2885	0.2563
application_type	BLA	1	0.1521	0.2311	0.4327	0.5106
year		1	0.0481	0.0417	1.3318	0.2485
regulatory_pathway	351(a)	1	0.0398	0.5443	0.0053	0.9417
regulatory_pathway	351(k)	1	1.4064	0.8646	2.6462	0.1038
regulatory_pathway	505(b)	1	-0.9345	0.4137	5.1041	0.0239
regulatory_pathway	Not Available	1	-0.6467	0.3490	3.4342	0.0639

Logistic Regression: Predicting CMC Deficiencies
Predict CMC with Multiple Predictors

Odds Ratio Estimates			
Effect	**Point Estimate**	**95% Wald Confidence Limits**	
application_type BLA vs NDA	1.355	0.548	3.354
year	1.049	0.967	1.139
regulatory_pathway 351(a) vs 505(b)(2)	0.909	0.212	3.906
regulatory_pathway 351(k) vs 505(b)(2)	3.566	0.383	33.214
regulatory_pathway 505(b) vs 505(b)(2)	0.343	0.171	0.687
regulatory_pathway Not Available vs 505(b)(2)	0.458	0.203	1.031

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	62.3	**Somers' D**	0.289
Percent Discordant	33.4	**Gamma**	0.302
Percent Tied	4.3	**Tau-a**	0.099
Pairs	27318	**c**	0.645

Model Information	
Data Set	WORK.DOMAIN_EVENT_MODEL2
Response Variable	multi_f
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	401
Number of Observations Used	401

Response Profile		
Ordered Value	**multi_f**	**Total Frequency**
1	1	160
2	0	241

Probability modeled is multi_f=1.

Class Level Information					
Class	**Value**	**Design Variables**			
application_type	**BLA**	1			
	NDA	0			
regulatory_pathway	**351(a)**	1	0	0	0
	351(k)	0	1	0	0

Logistic Regression: Predicting CMC Deficiencies Predict Multi-Domain CRL

Class Level Information					
Class	Value	Design Variables			
	505(b)	0	0	1	0
	505(b)(2)	0	0	0	0
	Not Available	0	0	0	1

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	541.429	501.244
SC	545.423	529.202
-2 Log L	539.429	487.244

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	52.1851	6	<.0001
Score	44.9387	6	<.0001
Wald	33.1952	6	<.0001

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
application_type	1	15.2385	<.0001
year	1	8.6363	0.0033
regulatory_pathway	4	21.4368	0.0003

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	227.9	77.5760	8.6338	0.0033
application_type	BLA	1	-2.9788	0.7631	15.2385	<.0001
year		1	-0.1129	0.0384	8.6363	0.0033
regulatory_pathway	351(a)	1	2.5307	0.8816	8.2397	0.0041
regulatory_pathway	351(k)	1	2.6738	0.8871	9.0850	0.0026

Logistic Regression: Predicting CMC Deficiencies Predict Multi-Domain CRL

Analysis of Maximum Likelihood Estimates						
Parameter		**DF**	**Estimate**	**Standard Error**	**Wald Chi-Square**	**Pr > ChiSq**
regulatory_pathway	**505(b)**	1	-1.1709	0.3840	9.2963	0.0023
regulatory_pathway	**Not Available**	1	0.6932	0.3597	3.7148	0.0539

Odds Ratio Estimates			
Effect	**Point Estimate**	**95% Wald Confidence Limits**	
application_type BLA vs NDA	0.051	0.011	0.227
year	0.893	0.828	0.963
regulatory_pathway 351(a) vs 505(b)(2)	12.563	2.232	70.719
regulatory_pathway 351(k) vs 505(b)(2)	14.494	2.548	82.467
regulatory_pathway 505(b) vs 505(b)(2)	0.310	0.146	0.658
regulatory_pathway Not Available vs 505(b)(2)	2.000	0.988	4.048

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	64.7	**Somers' D**	0.336
Percent Discordant	31.1	**Gamma**	0.351
Percent Tied	4.1	**Tau-a**	0.162
Pairs	38560	**c**	0.668

Model Information	
Data Set	SASDATA.DOMAIN_EVENT_MODEL
Response Variable	cmc_f
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	401
Number of Observations Used	401

Response Profile		
Ordered Value	**cmc_f**	**Total Frequency**
1	1	314
2	0	87

Probability modeled is cmc_f=1.

Logistic Regression: Predicting CMC Deficiencies Trend Over Time

Class Level Information		
Class	Value	Design Variables
application_type	BLA	1
	NDA	0

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	421.470	421.643
SC	425.464	433.625
-2 Log L	419.470	415.643

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	3.8272	2	0.1476
Score	3.5925	2	0.1659
Wald	3.5198	2	0.1721

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
year	1	0.0424	0.8369
application_type	1	3.0352	0.0815

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-12.3202	65.4697	0.0354	0.8507
year		1	0.00667	0.0324	0.0424	0.8369
application_type	BLA	1	0.5767	0.3310	3.0352	0.0815

Logistic Regression: Predicting CMC Deficiencies Trend Over Time

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
year	1.007	0.945	1.073
application_type BLA vs NDA	1.780	0.930	3.406

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	50.5	Somers' D	0.069
Percent Discordant	43.6	Gamma	0.073
Percent Tied	5.8	Tau-a	0.023
Pairs	27318	c	0.534

Model Information	
Data Set	SASDATA.DOMAIN_EVENT_MODEL
Distribution	Binomial
Link Function	Logit
Dependent Variable	cmc_f

Number of Observations Read	401
Number of Observations Used	401
Number of Events	314
Number of Trials	401

Class Level Information		
Class	Value	Design Variables
application_type	BLA	1
	NDA	0

Response Profile		
Ordered Value	cmc_f	Total Frequency
1	1	314
2	0	87

PROC GENMOD is modeling the probability that cmc_f='1'.

Logistic Regression: Predicting CMC Deficiencies
Clustered Sensitivity Model: CMC Deficiency by Application Type

Parameter Information		
Parameter	**Effect**	**application_type**
Prm1	Intercept	
Prm2	application_type	BLA

Algorithm converged.

GEE Model Information	
Correlation Structure	Independent
Subject Effect	application_number (298 levels)
Number of Clusters	298
Correlation Matrix Dimension	6
Maximum Cluster Size	6
Minimum Cluster Size	1

Algorithm converged.

GEE Fit Criteria	
QIC	420.8517
QICu	419.6853

Analysis Of GEE Parameter Estimates							
Empirical Standard Error Estimates							
Parameter		**Estimate**	**Standard Error**	**95% Confidence Limits**		**Z**	**Pr > |Z|**
Intercept		1.1606	0.1539	0.8590	1.4622	7.54	<.0001
application_type	BLA	0.5948	0.3603	-0.1114	1.3010	1.65	0.0988

Contrast Estimate Results										
		Mean					**L'Beta**			
Label	**Mean Estimate**	**Confidence Limits**		**L'Beta Estimate**	**Standard Error**	**Alpha**	**Confidence Limits**		**Chi-Square**	**Pr > ChiSq**
BLA vs NDA	0.6445	0.4722	0.7860	0.5948	0.3603	0.05	-0.1114	1.3010	2.73	0.0988
Exp(BLA vs NDA)				1.8127	0.6531	0.05	0.8946	3.6730		

Logistic Regression: Predicting CMC Deficiencies
Clustered Sensitivity Model: CMC Deficiency by Application Type and Year

Model Information	
Data Set	SASDATA.DOMAIN_EVENT_MODEL
Distribution	Binomial
Link Function	Logit
Dependent Variable	cmc_f

Number of Observations Read	401
Number of Observations Used	401
Number of Events	314
Number of Trials	401

Class Level Information		
Class	**Value**	**Design Variables**
application_type	**BLA**	1
	NDA	0

Response Profile		
Ordered Value	**cmc_f**	**Total Frequency**
1	1	314
2	0	87

PROC GENMOD is modeling the probability that cmc_f='1'.

Parameter Information		
Parameter	**Effect**	**application_type**
Prm1	Intercept	
Prm2	application_type	BLA
Prm3	year	

Algorithm converged.

GEE Model Information	
Correlation Structure	Independent
Subject Effect	application_number (298 levels)
Number of Clusters	298
Correlation Matrix Dimension	6

Logistic Regression: Predicting CMC Deficiencies
Clustered Sensitivity Model: CMC Deficiency by Application Type and Year

GEE Model Information	
Maximum Cluster Size	6
Minimum Cluster Size	1

Algorithm converged.

GEE Fit Criteria	
QIC	424.4054
QICu	421.6431

Analysis Of GEE Parameter Estimates							
Empirical Standard Error Estimates							
Parameter		**Estimate**	**Standard Error**	**95% Confidence Limits**		**Z**	**Pr > |Z|**
Intercept		-12.3202	87.8562	-184.515	159.8749	-0.14	0.8885
application_type	BLA	0.5767	0.3661	-0.1409	1.2943	1.58	0.1152
year		0.0067	0.0435	-0.0786	0.0919	0.15	0.8780

Contrast Estimate Results										
		Mean					**L'Beta**			
Label	**Mean Estimate**	**Confidence Limits**		**L'Beta Estimate**	**Standard Error**	**Alpha**	**Confidence Limits**		**Chi-Square**	**Pr > ChiSq**
BLA vs NDA	0.6403	0.4648	0.7849	0.5767	0.3661	0.05	-0.1409	1.2943	2.48	0.1152
Exp(BLA vs NDA)				1.7802	0.6518	0.05	0.8686	3.6485		

Model Information	
Data Set	WORK.DOMAIN_EVENT_MODEL2
Distribution	Binomial
Link Function	Logit
Dependent Variable	multi_f

Number of Observations Read	401
Number of Observations Used	401
Number of Events	160
Number of Trials	401

Logistic Regression: Predicting CMC Deficiencies
Clustered Sensitivity Model: Multi-Domain CRL by Application Type and Year

Class Level Information		
Class	**Value**	**Design Variables**
application_type	**BLA**	1
	NDA	0

Response Profile		
Ordered Value	**multi_f**	**Total Frequency**
1	1	160
2	0	241

PROC GENMOD is modeling the probability that multi_f='1'.

Parameter Information		
Parameter	**Effect**	**application_type**
Prm1	Intercept	
Prm2	application_type	BLA
Prm3	year	

Algorithm converged.

GEE Model Information	
Correlation Structure	Independent
Subject Effect	application_number (298 levels)
Number of Clusters	298
Correlation Matrix Dimension	6
Maximum Cluster Size	6
Minimum Cluster Size	1

Algorithm converged.

GEE Fit Criteria	
QIC	521.6526
QICu	520.3824

Logistic Regression: Predicting CMC Deficiencies
Clustered Sensitivity Model: Multi-Domain CRL by Application Type and Year

Analysis Of GEE Parameter Estimates							
Empirical Standard Error Estimates							
Parameter		Estimate	Standard Error	95% Confidence Limits		Z	Pr > |Z|
Intercept		132.0322	65.7238	3.2159	260.8485	2.01	0.0445
application_type	BLA	-0.9810	0.3067	-1.5820	-0.3799	-3.20	0.0014
year		-0.0655	0.0325	-0.1292	-0.0017	-2.01	0.0443

Contrast Estimate Results										
Label	Mean Estimate	Mean Confidence Limits		L'Beta Estimate	Standard Error	Alpha	L'Beta Confidence Limits		Chi-Square	Pr > ChiSq
BLA vs NDA	0.2727	0.1705	0.4061	-0.9810	0.3067	0.05	-1.5820	-0.3799	10.23	0.0014
Exp(BLA vs NDA)				0.3749	0.1150	0.05	0.2056	0.6839		